

5G: Architecture and Key Principles

5G CORE - INTRODUCTION



Services

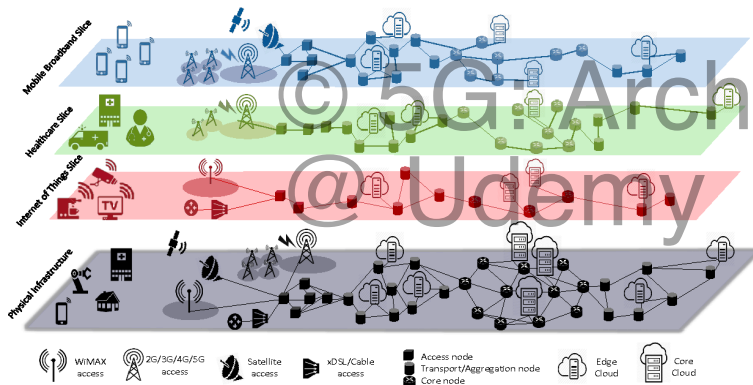
- LTE
 - Voice
 - Mobile broadband
- 5G
 - Voice
 - Mobile broadband
 - FWA
 - MMTC
 - URLLC

One Network
Many services

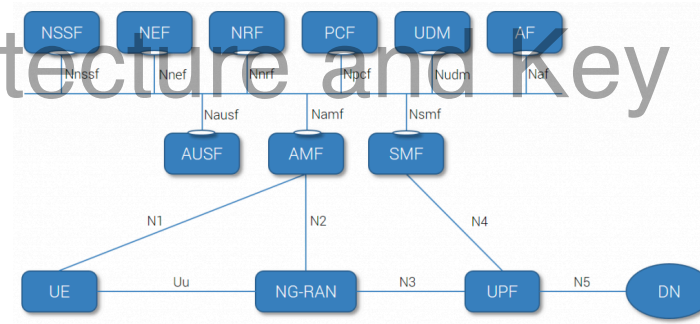
- Flexible
- Complex



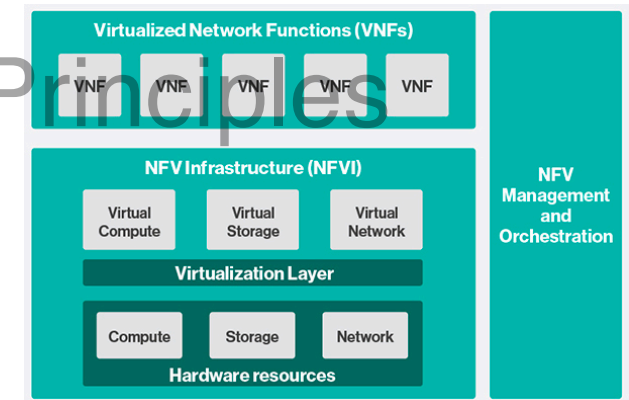
Key technologies



Network Slicing



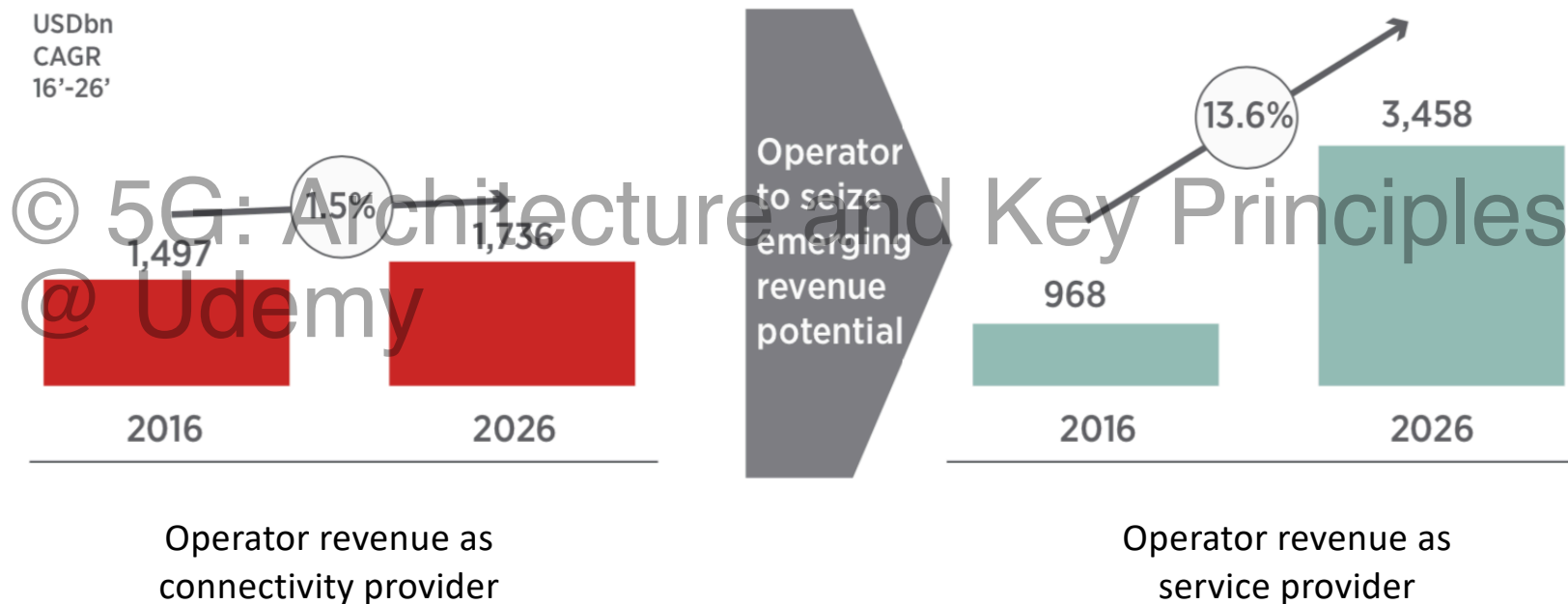
Service Based Architecture



Virtualization



Business Landscape

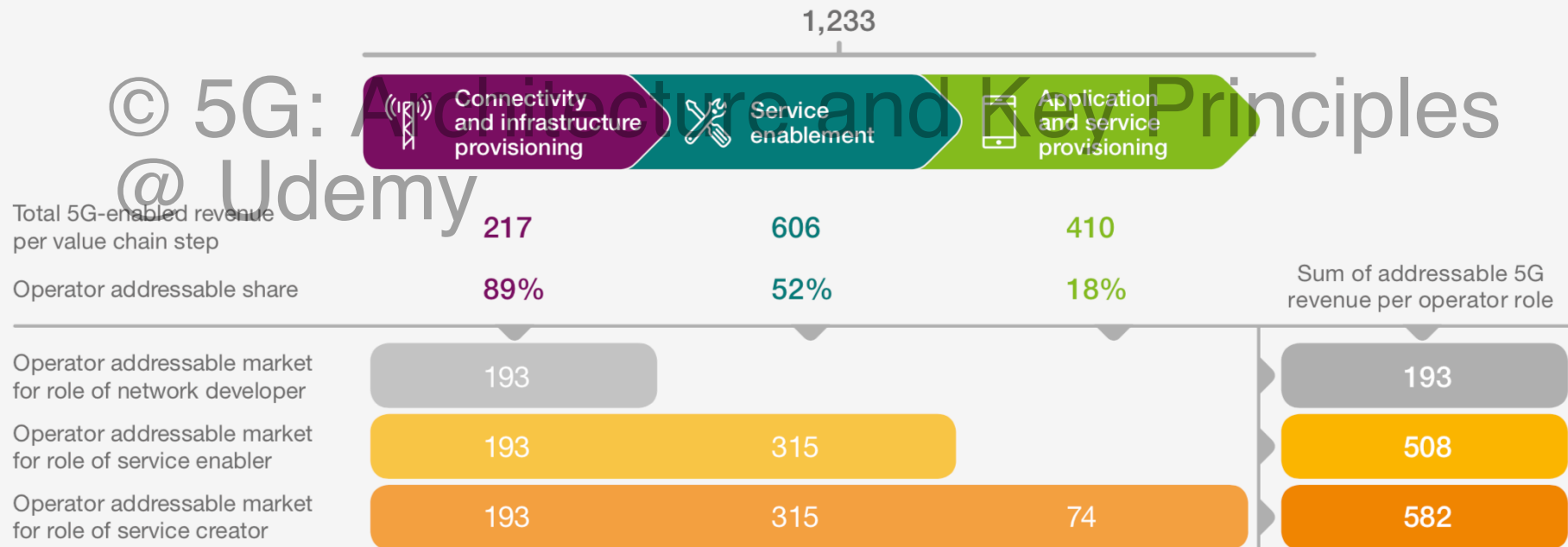


Operators can win bigger business as service providers



Business Landscape

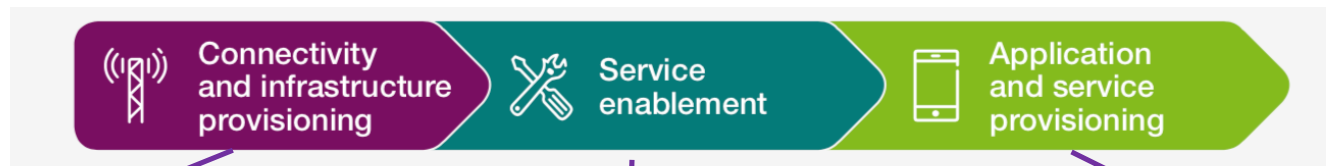
Figure 3: Operator addressable 5G market in eight industries (USD billion)



Source: Ericsson and Arthur D. Little



Operators' role in 5G



Connectivity provider:
Operators provide telecom infrastructure

Example:

- Operator as connectivity provider
- Acamai provides CDN outside operator's network
- Netflix as video-streaming service provider

Service Enabler:

Expose network capabilities for other service providers

Example:

- Operator as connectivity provider
- Operator's network as CDN
- Netflix as video-streaming service provider

Service provider:

Provide service to the end user directly

Example:

- Operator as connectivity provider
- Operator's network as CDN
- Operator as video-streaming service provider



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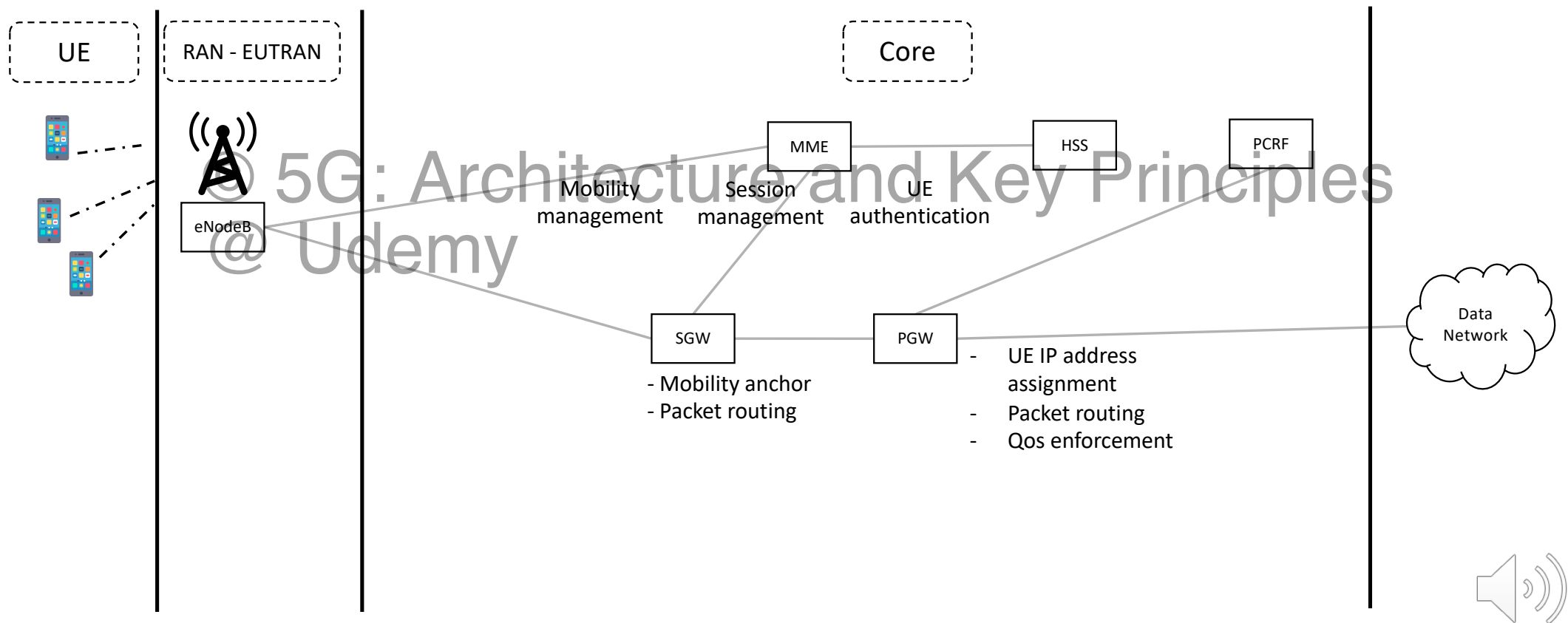


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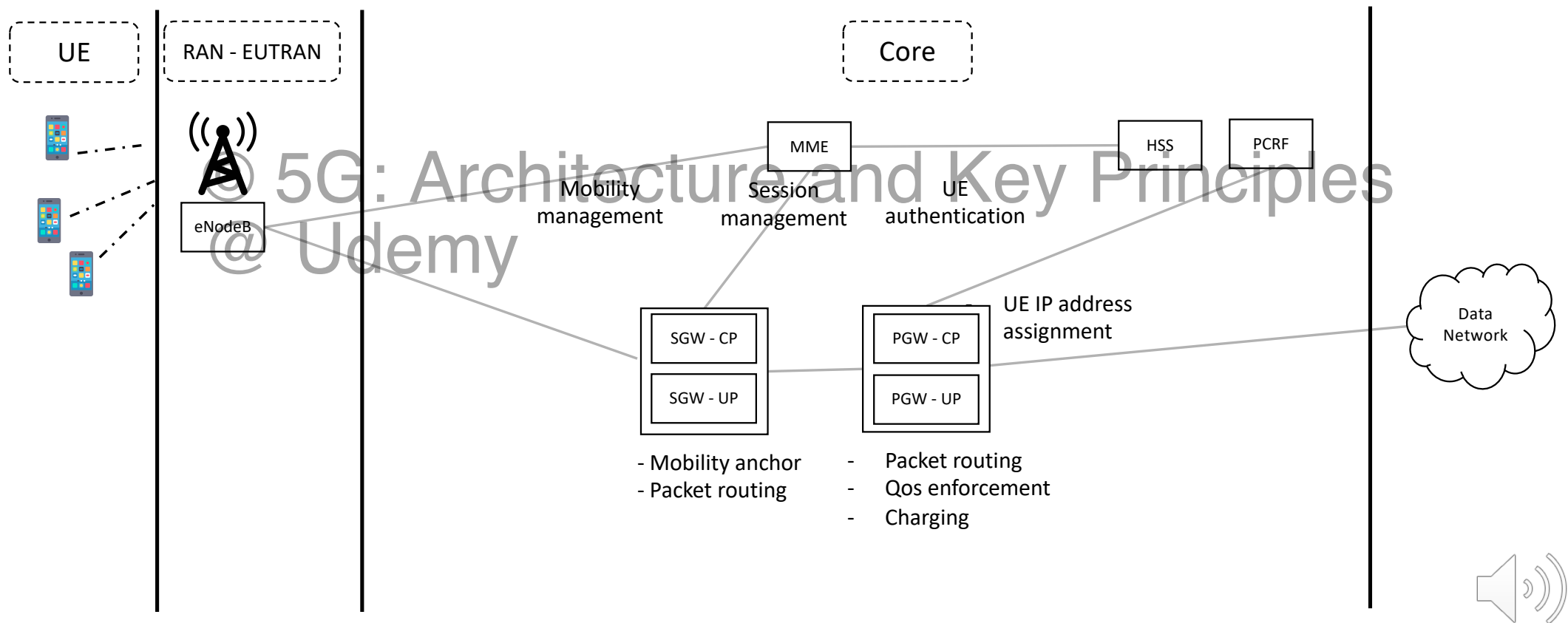
5G CORE



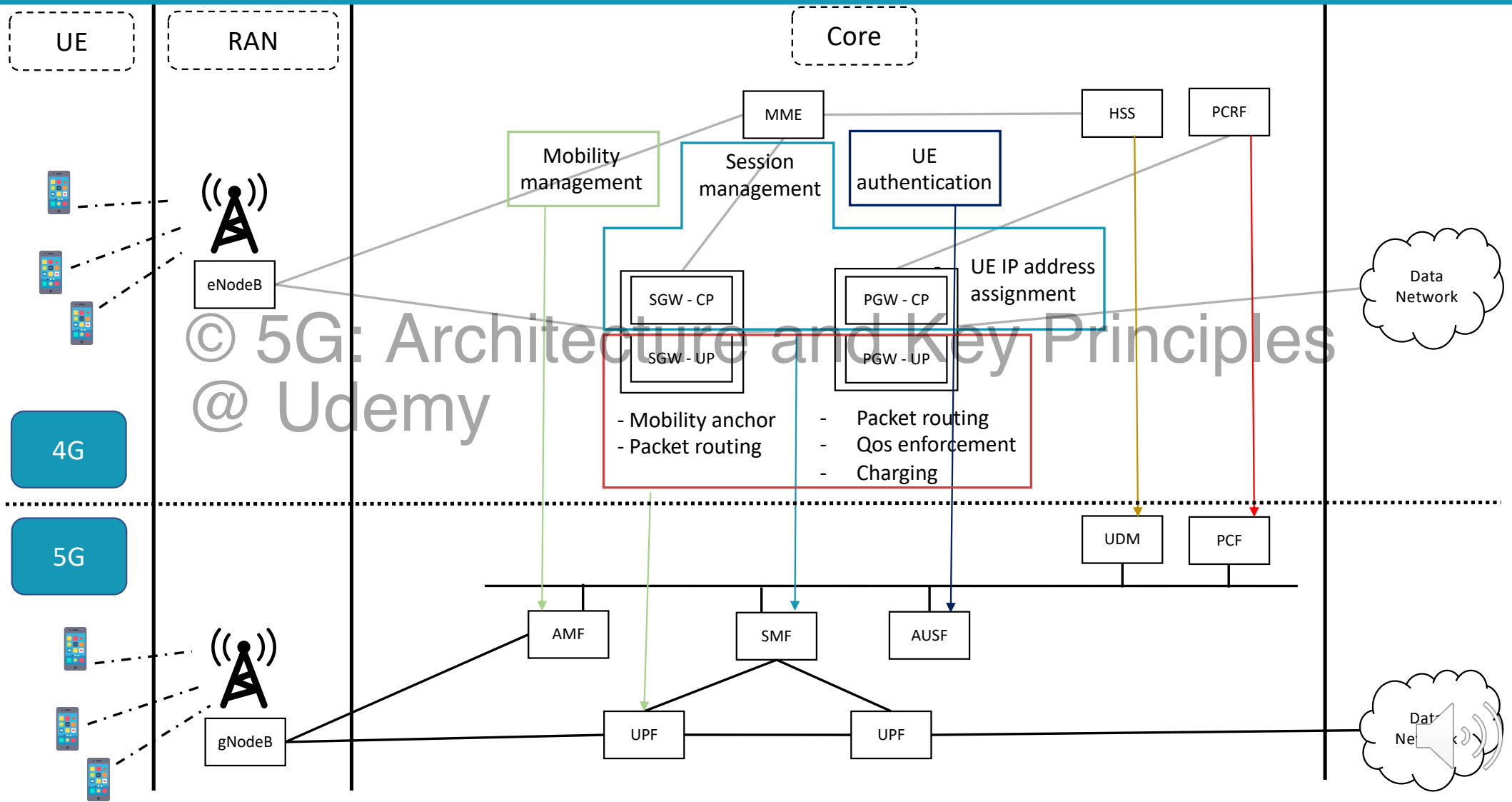
4G Network



4G Network



4G and 5G Network



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NETWORK SLICING

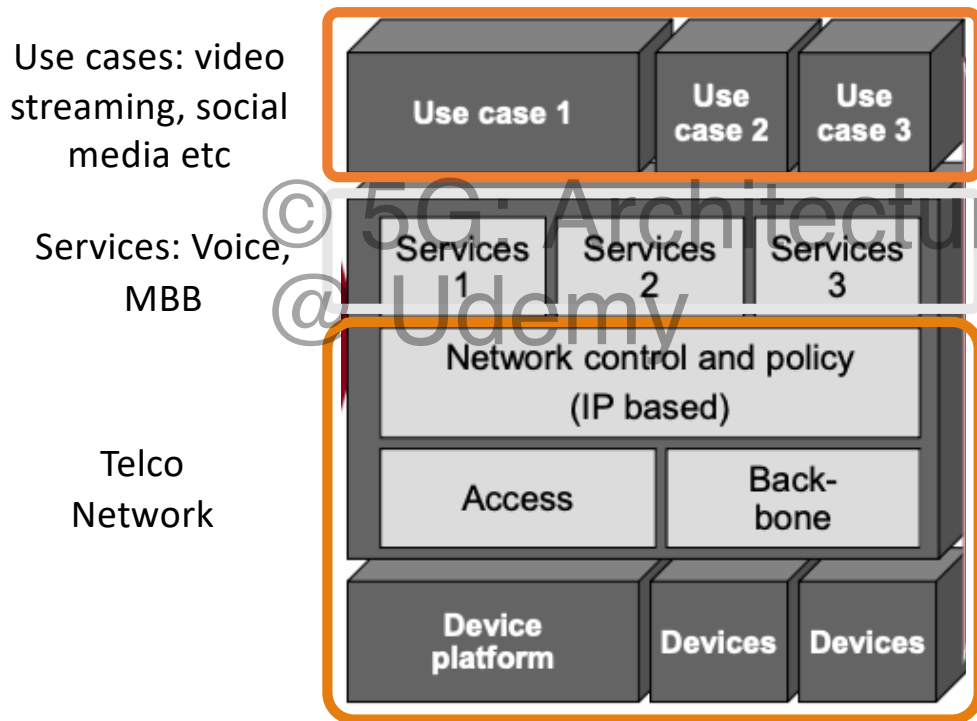
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Principles

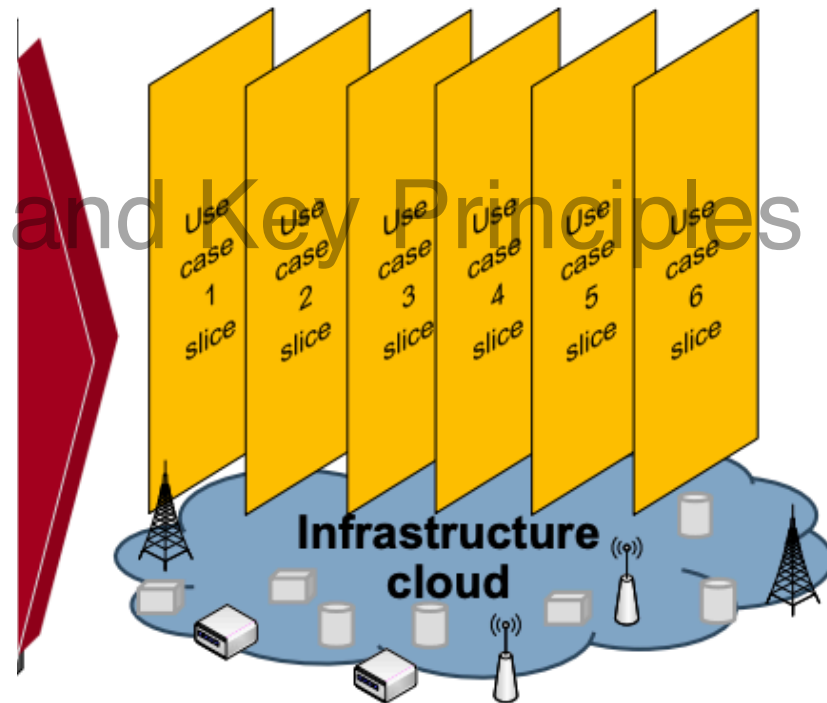
NETWORK SLICING



Network Slicing



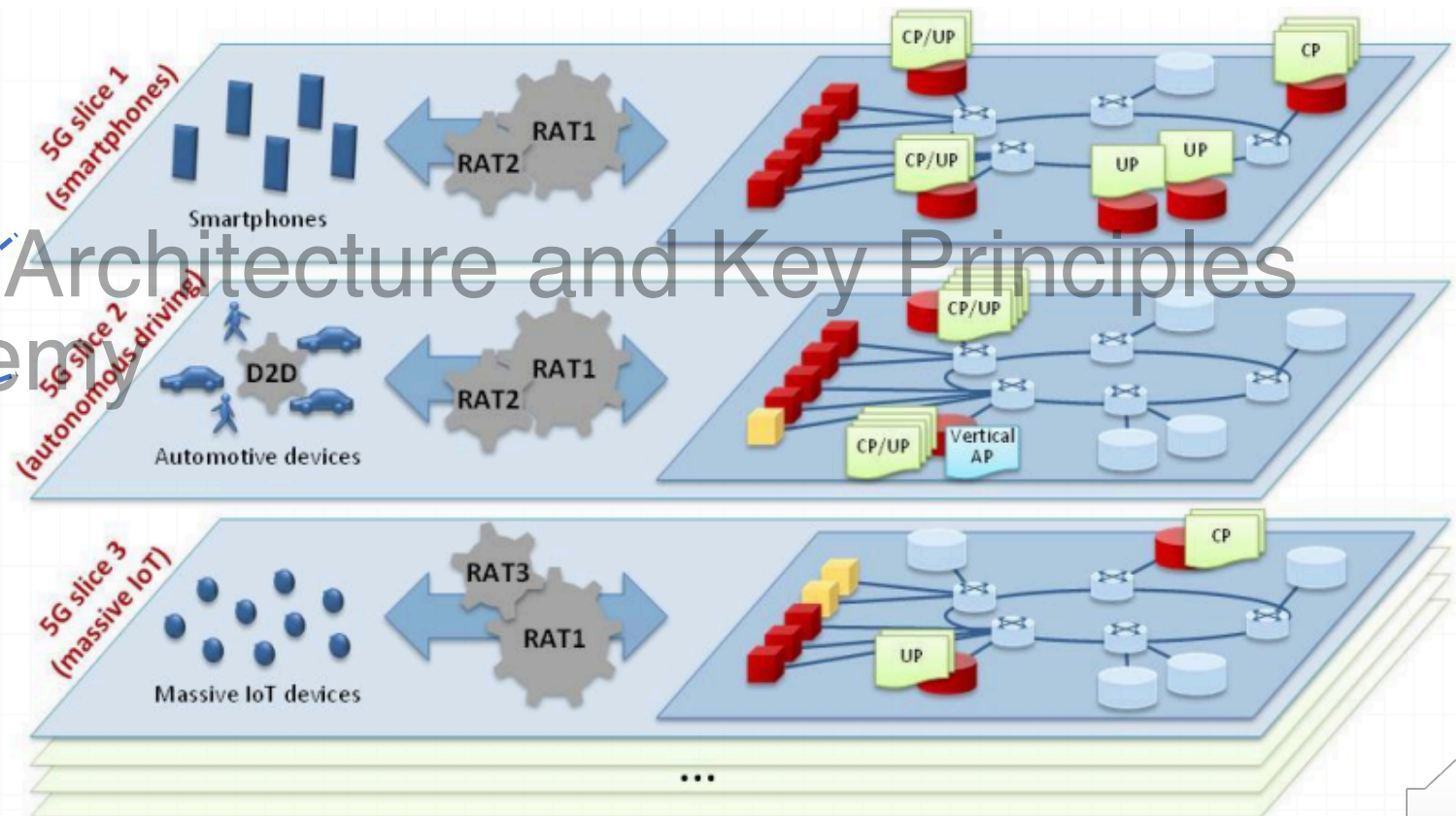
Present



Future

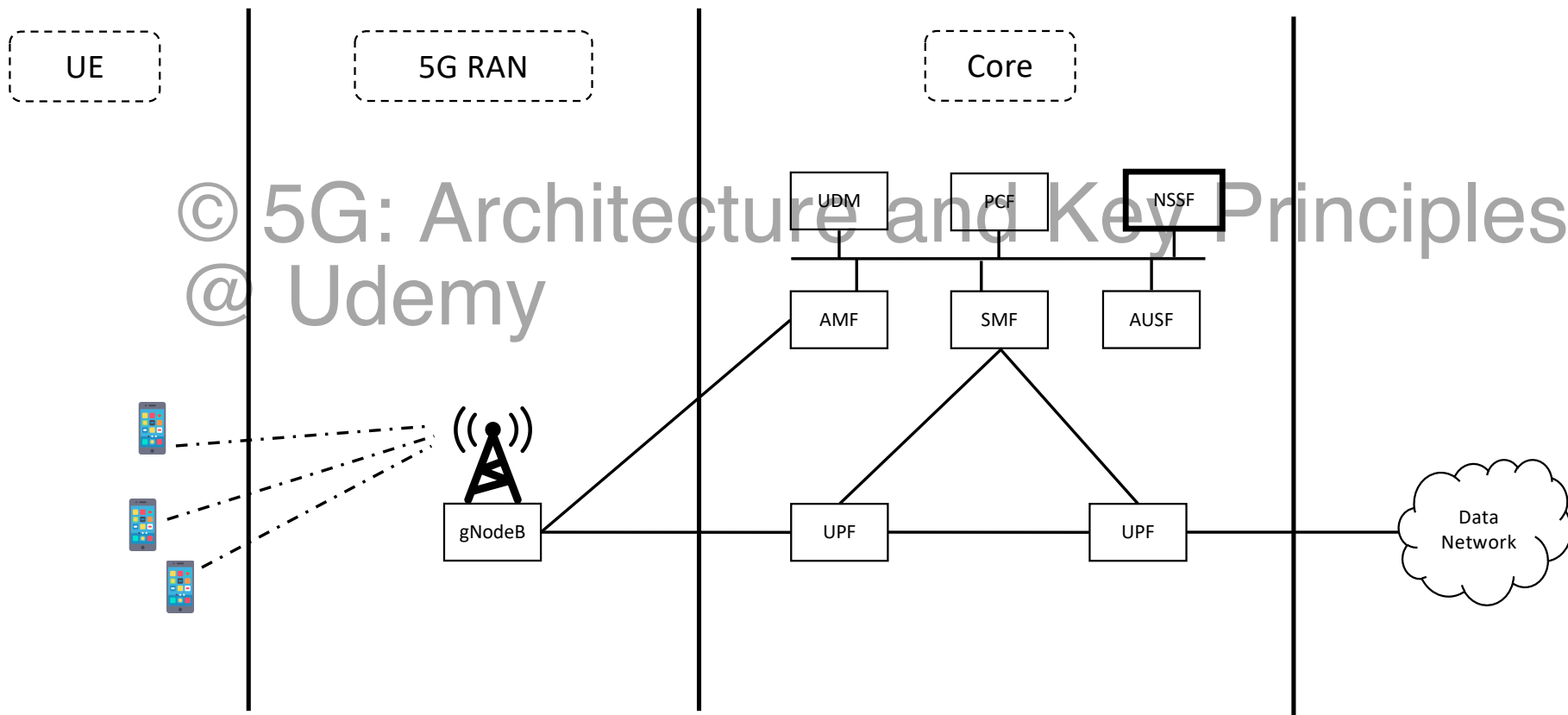


Network Slicing



RAT= Radio Access Technology; CP = Control Plane; UP= User Plane; AP= Access Point; IoT= Internet of things; D2D = Device to Device

Network Slice Selection Function (NSSF)



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SERVICE BASED
ARCHITECTURE



Monolithic architecture

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- Self contained
- All functions delivered and hosted together
- Failure handling
- Hard to scale

All functions are built as
one block of software



Monolithic architecture - Example

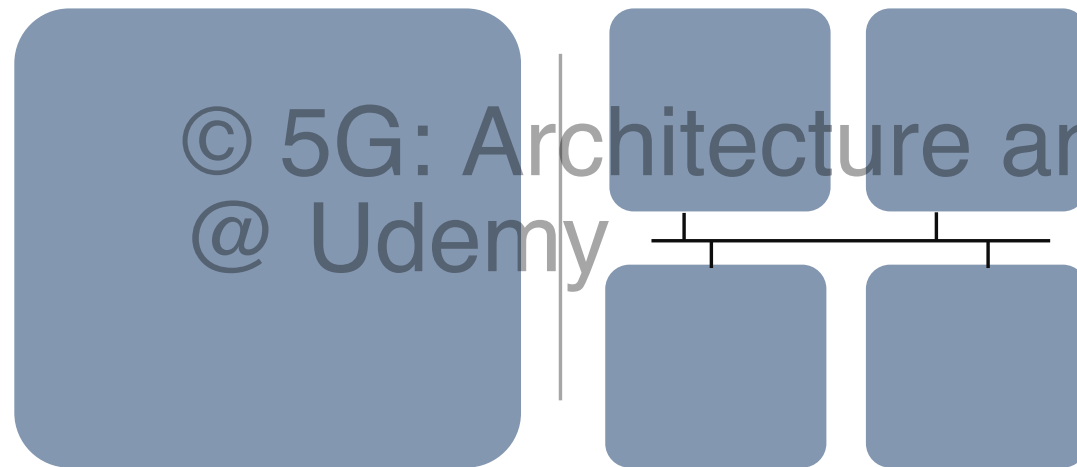


Monolithic architecture

- Example: E-Commerce website



Service based architecture (SBA)



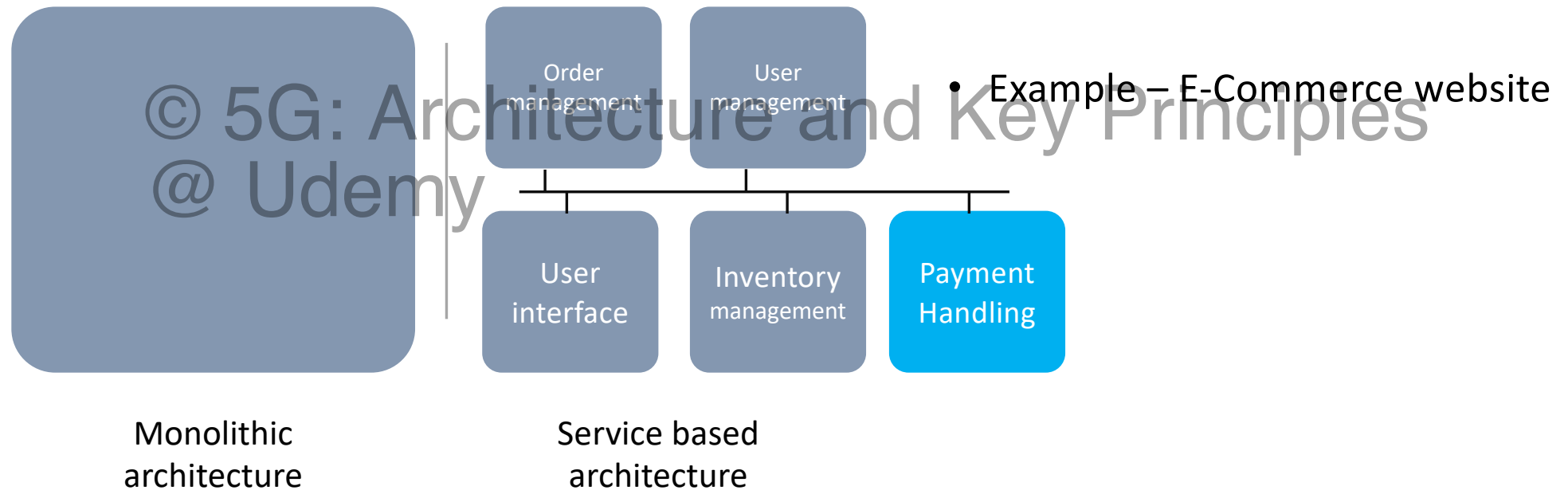
Monolithic
architecture

Service based
architecture

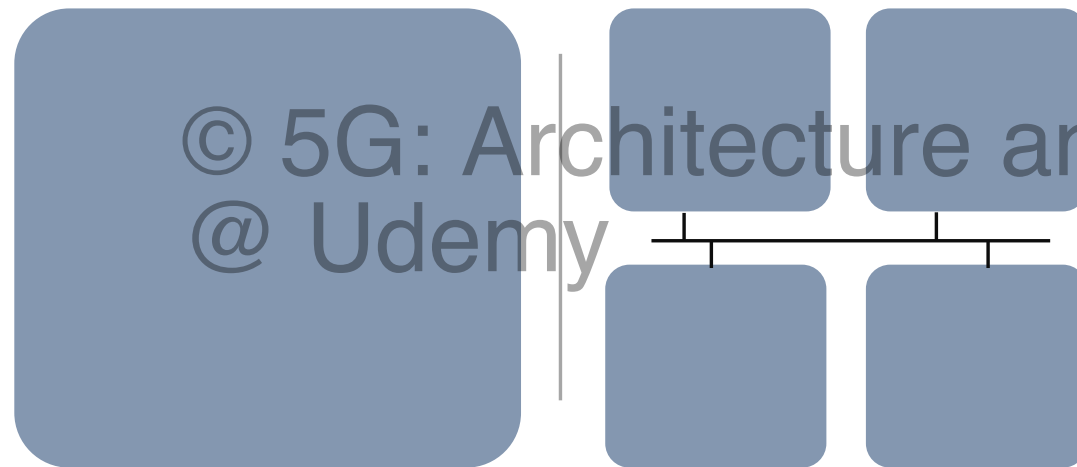
- A set of communicating services
- Flexible development



Service based architecture (SBA) - Example



Service based architecture (SBA)



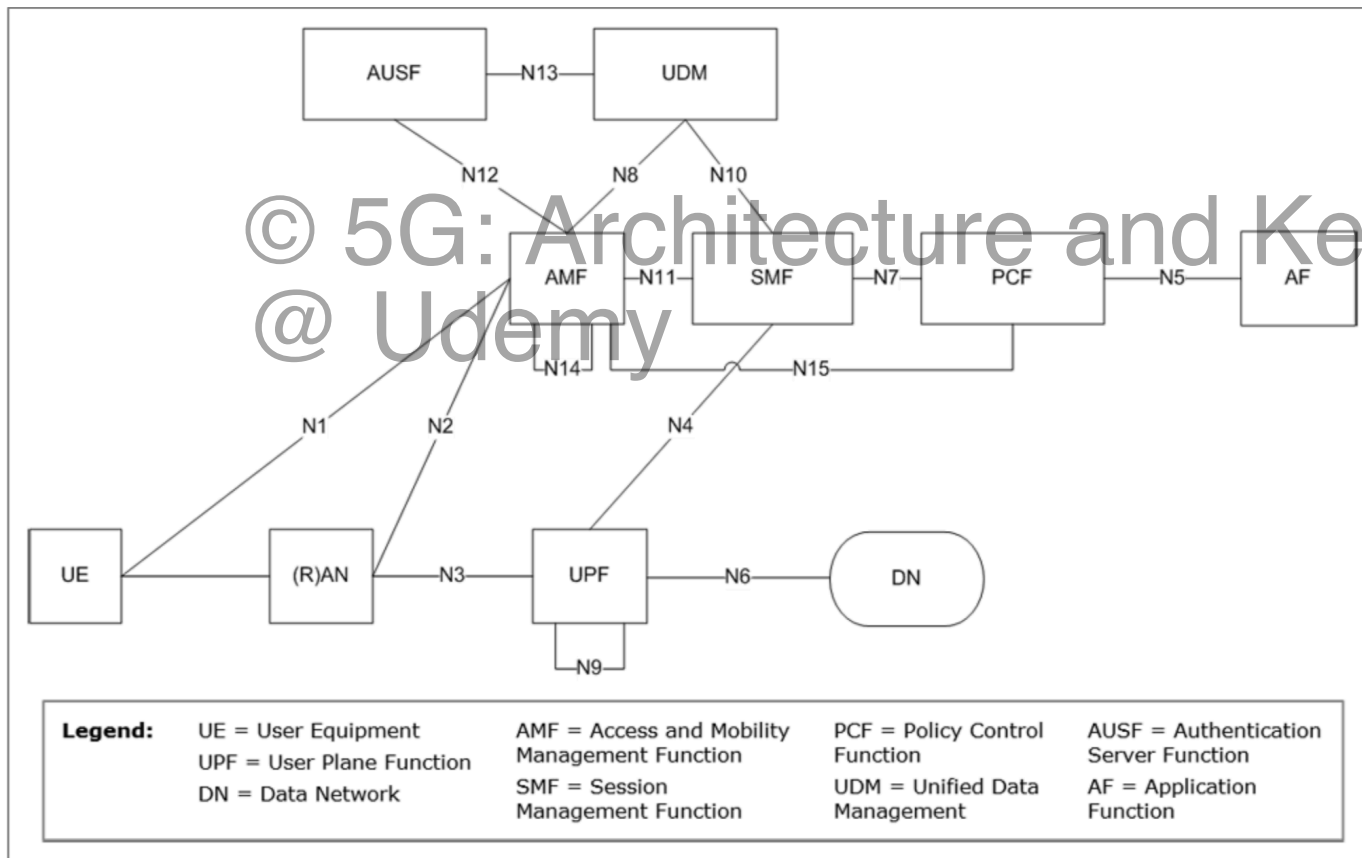
Monolithic
architecture

Service based
architecture

- Web based interface
- Development support



5G Core – Point to point architecture

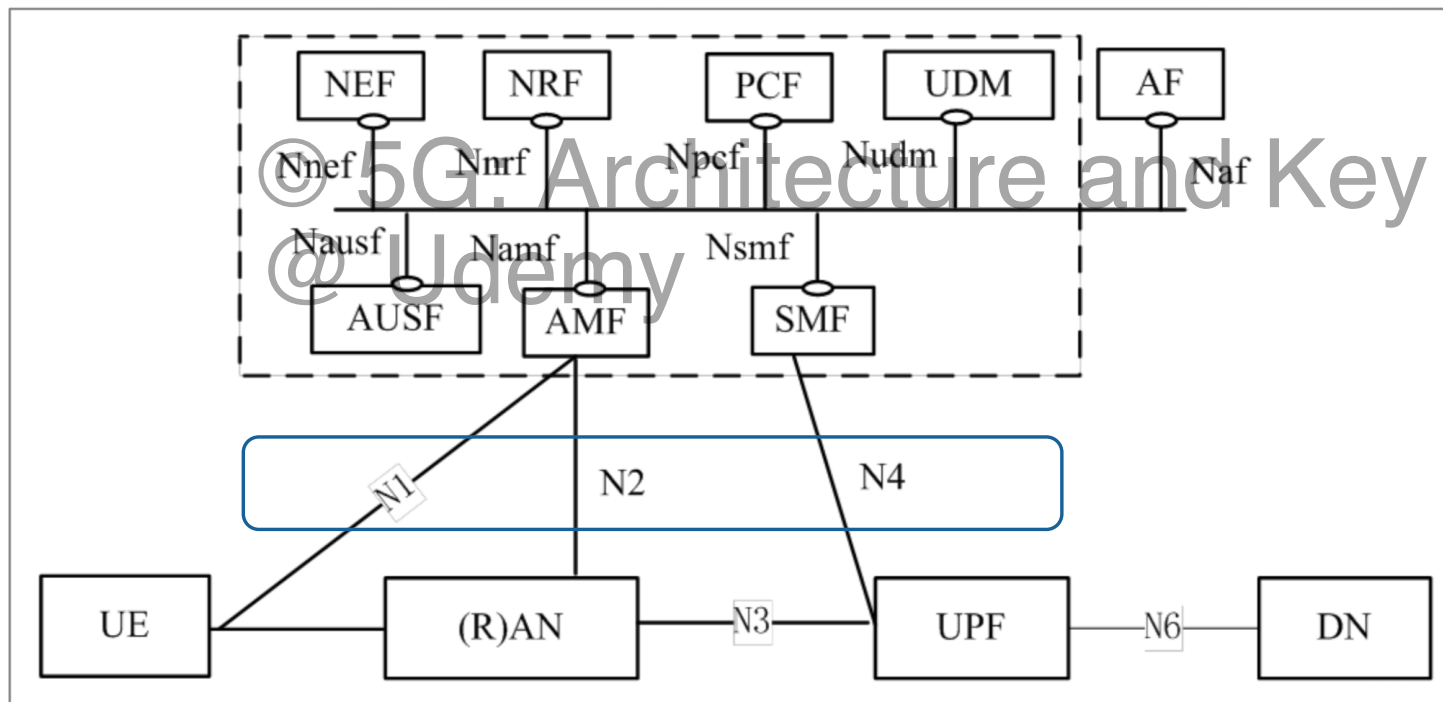


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Complex to add
new functions



5G Core – Service based architecture



Principles

SBA opens the control plane for innovation



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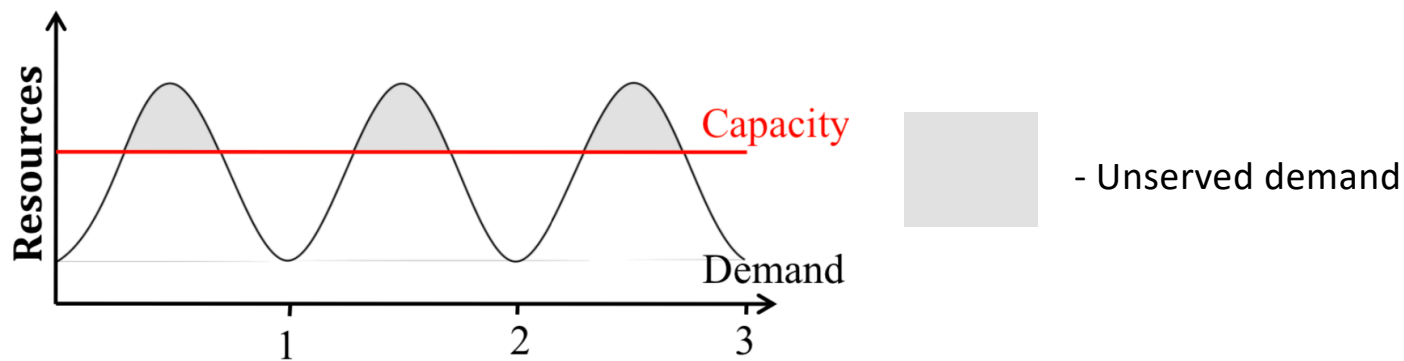
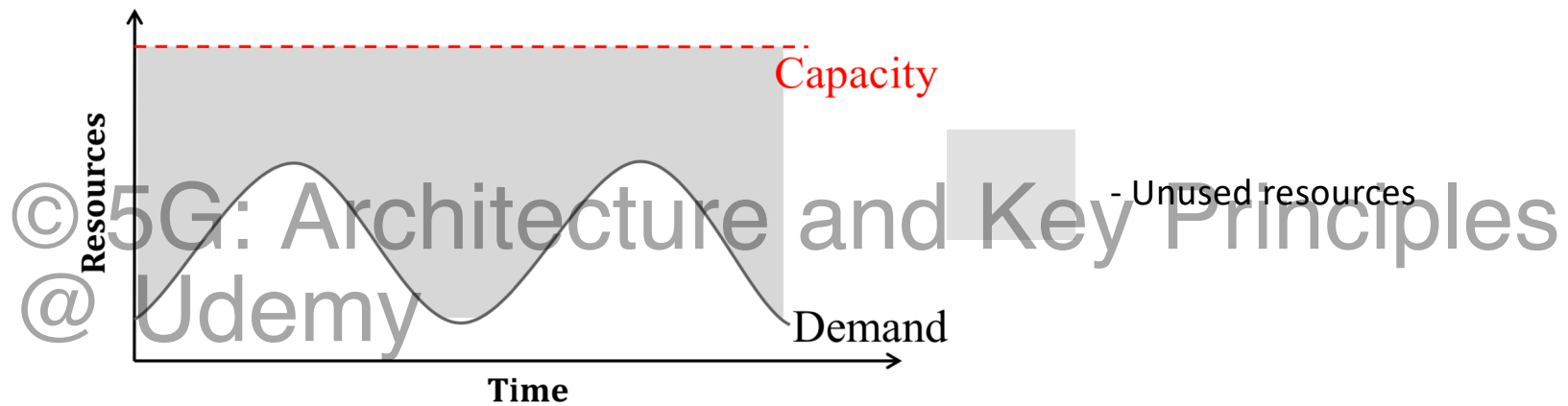


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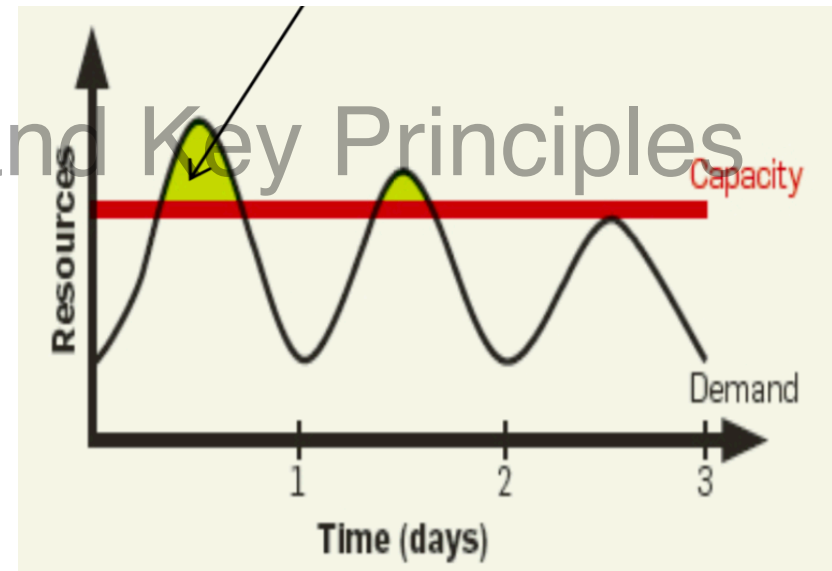
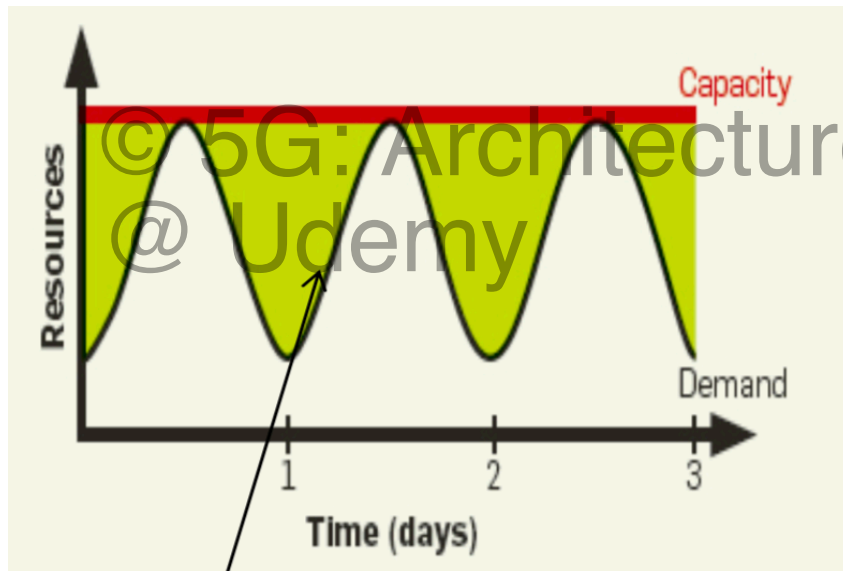
CLOUD COMPUTING AND
VIRTUALIZATION



Server based systems



Server based systems



"Average server utilization is 5% to 20%"



Cloud computing

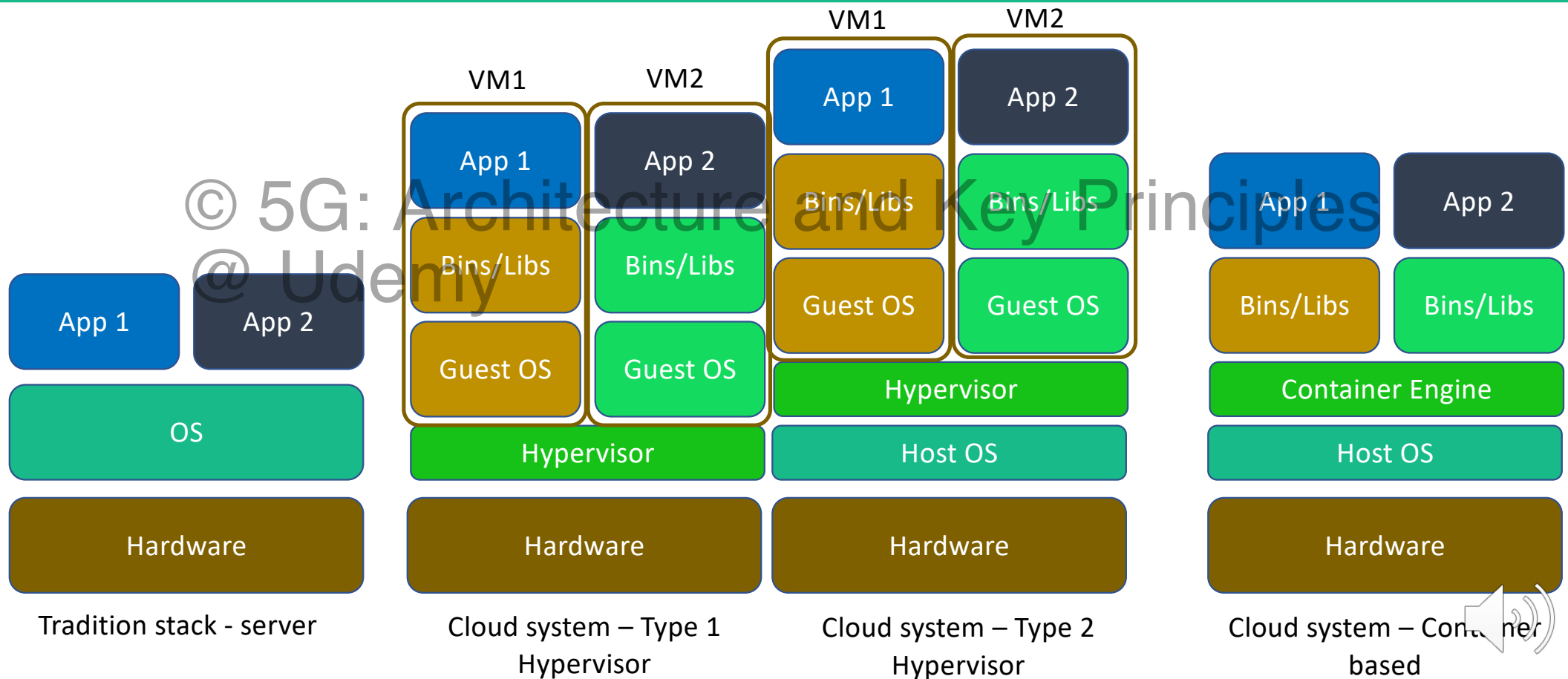
"on-demand availability of computer system resources, especially data storage and computing power, without direct active management by the user."

Example: Gmail Enterprise

- On demand storage
- On demand user accounts



Cloud computing



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NETWORK FUNCTION
VIRTUALIZATION (NFV)



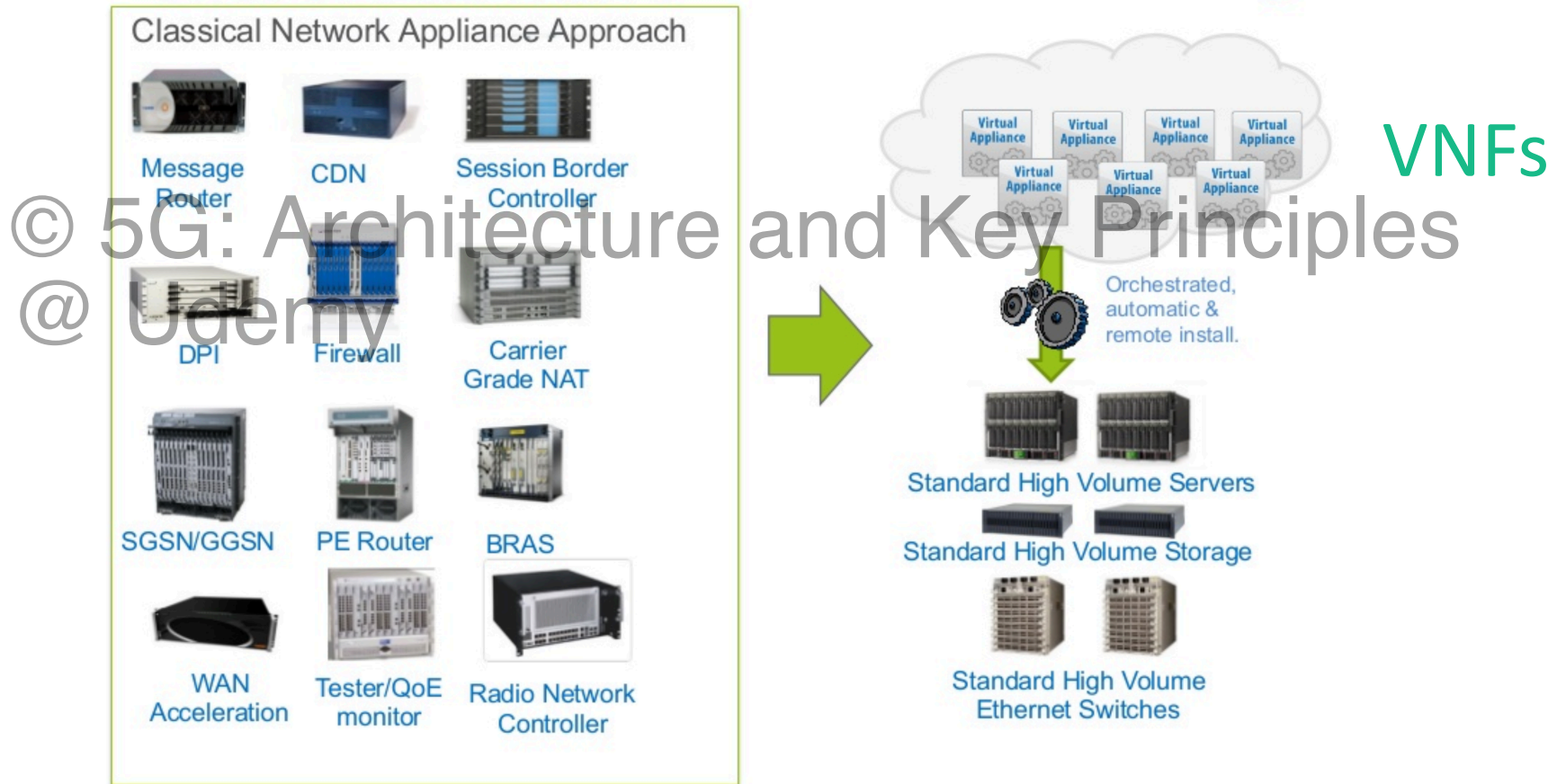
NFV

Purpose Built Hardware  Software based devices

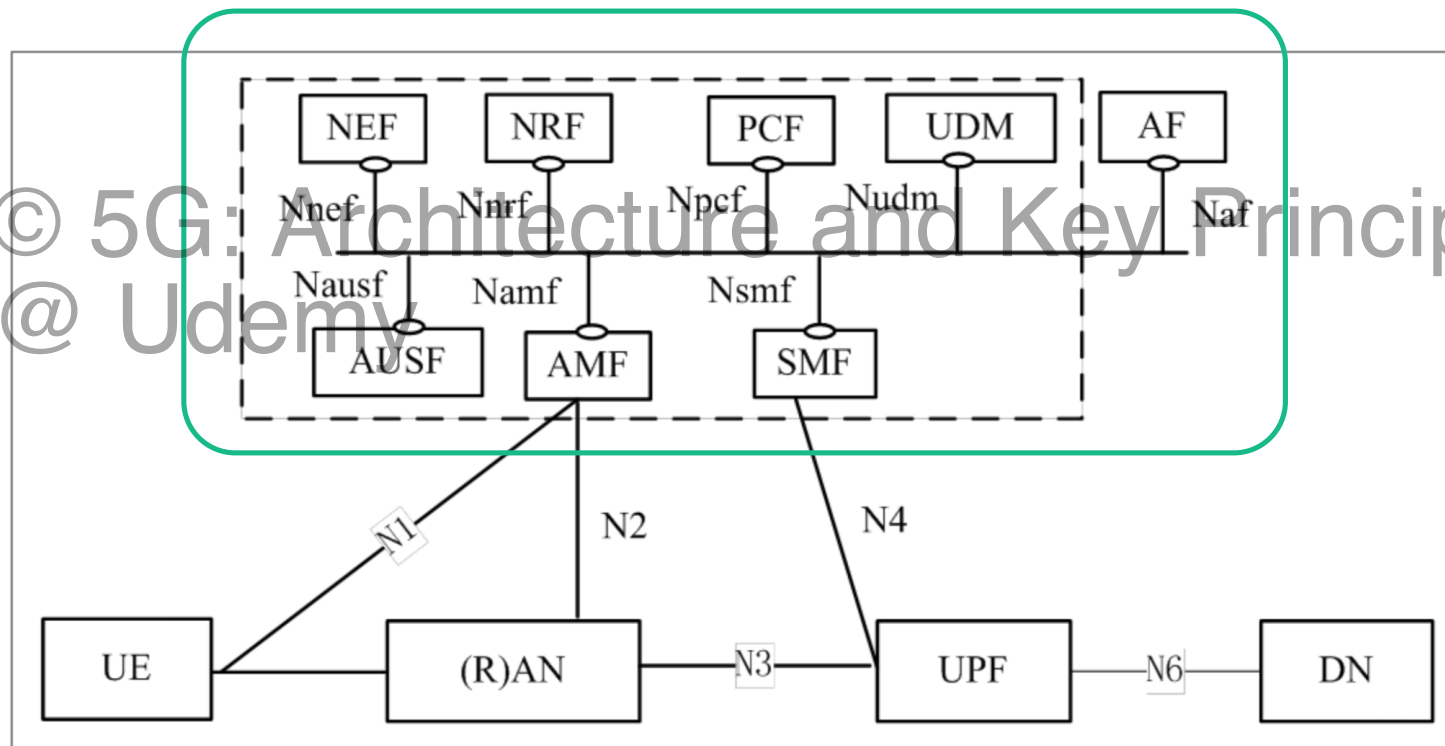
"NFV is this process of decoupling the network functions from proprietary hardware appliances so they can run in software on standardized hardware."



Virtualized Network Function (VNF)



VNFs in 5G core



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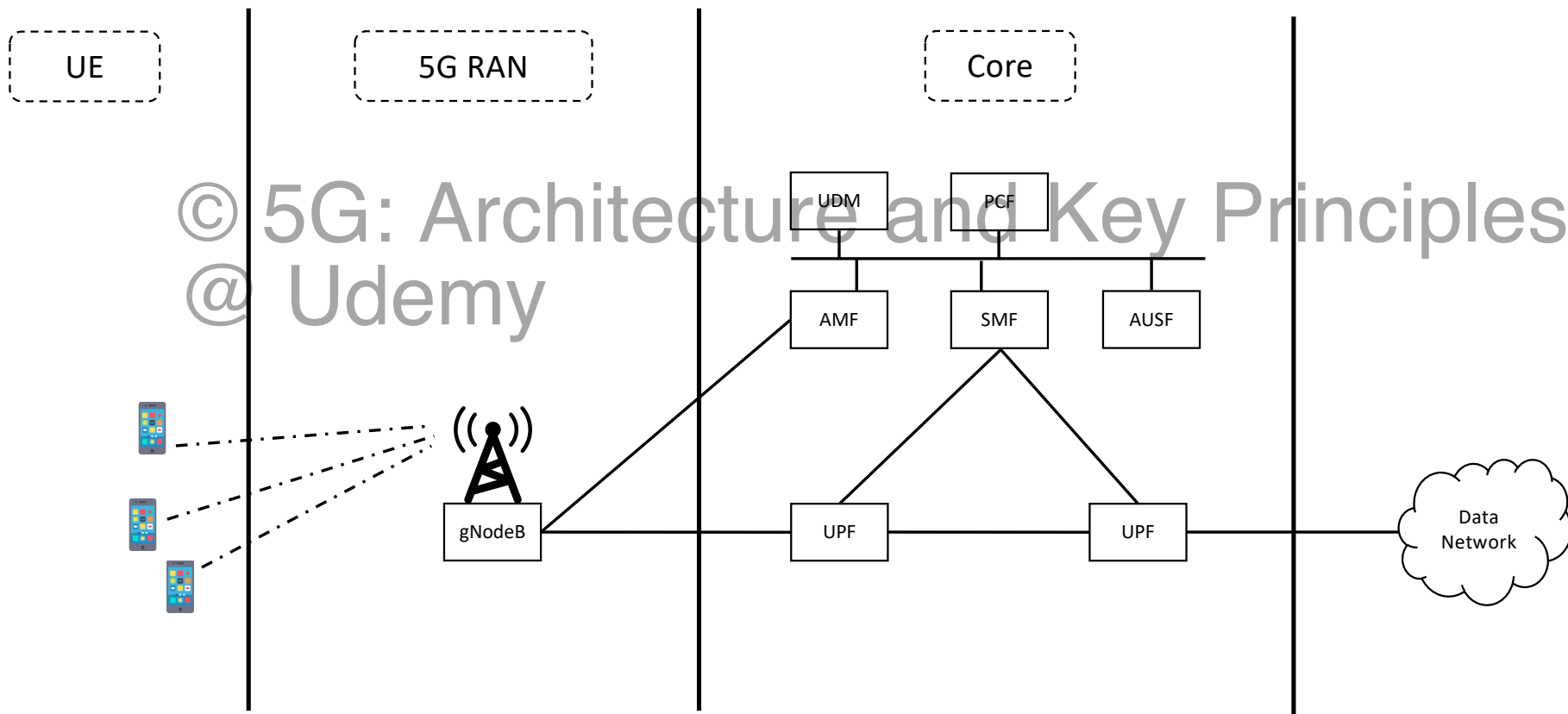
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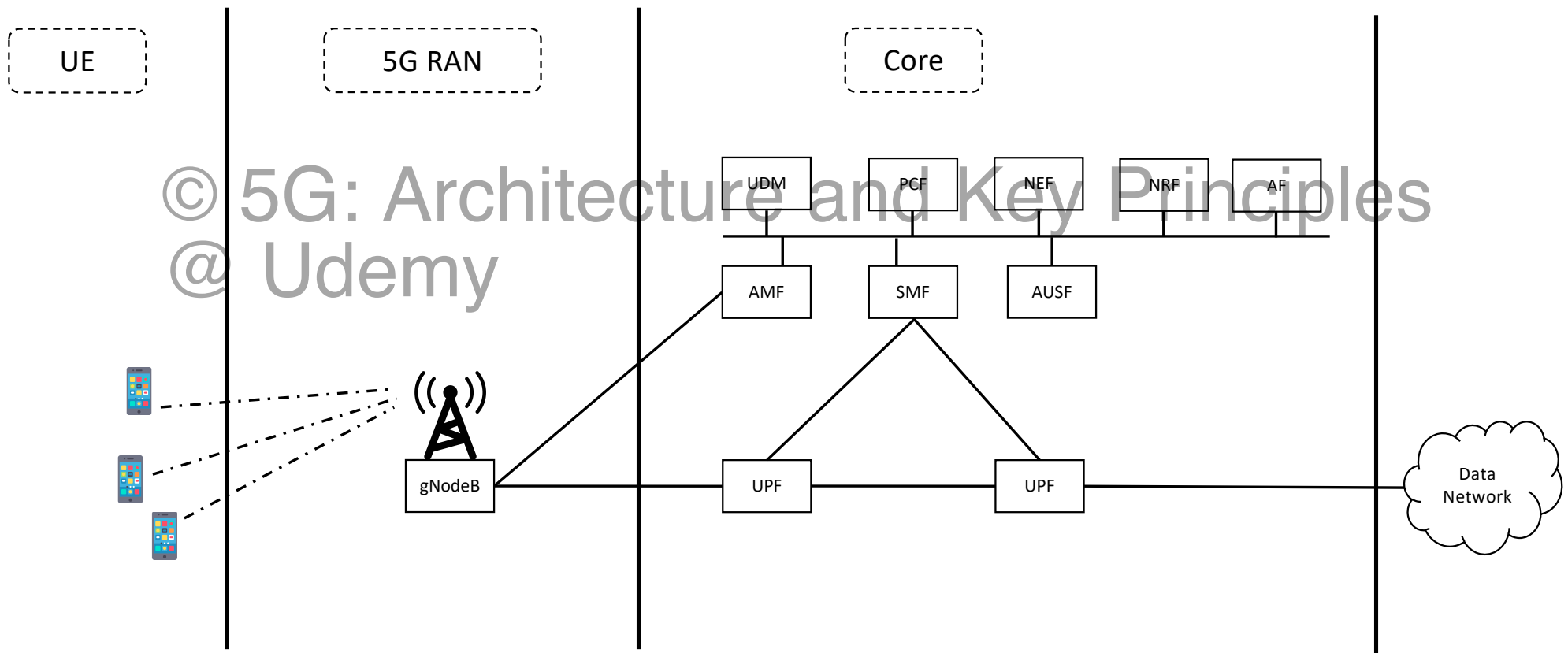
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CORE NETWORK FUNCTIONS

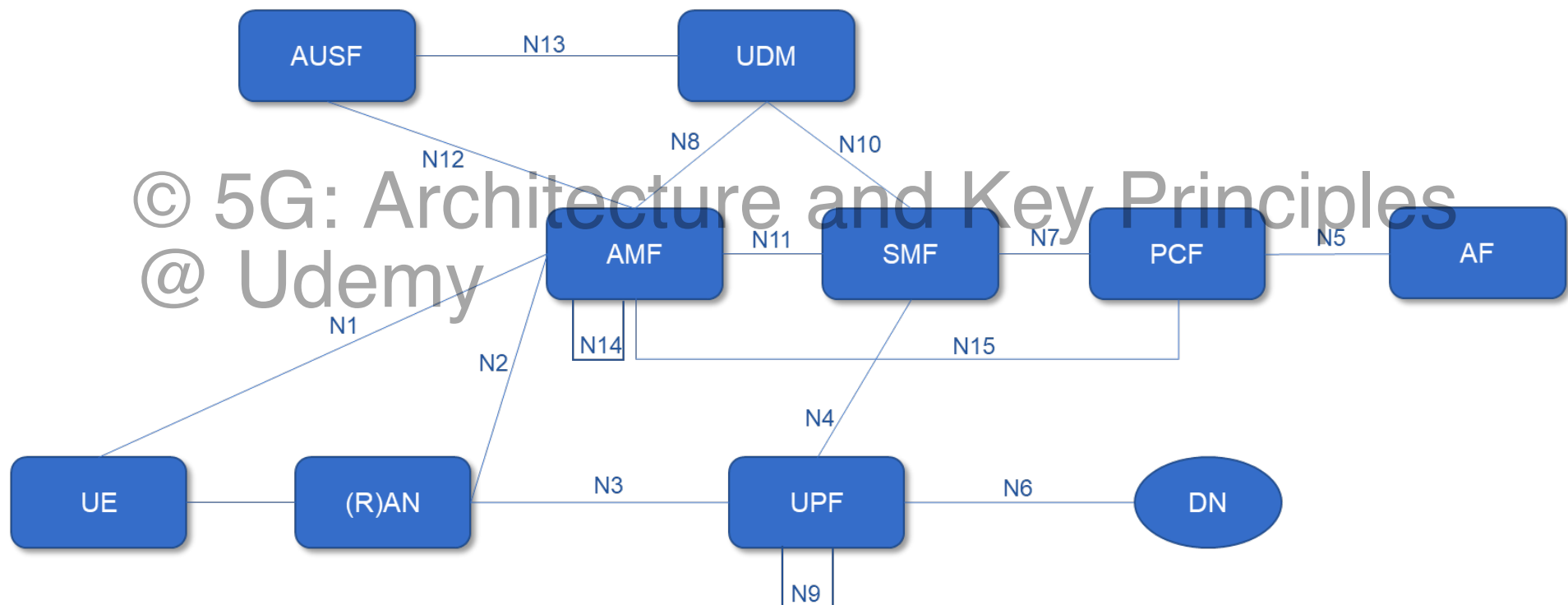
Core Network Functions



Core Network Functions

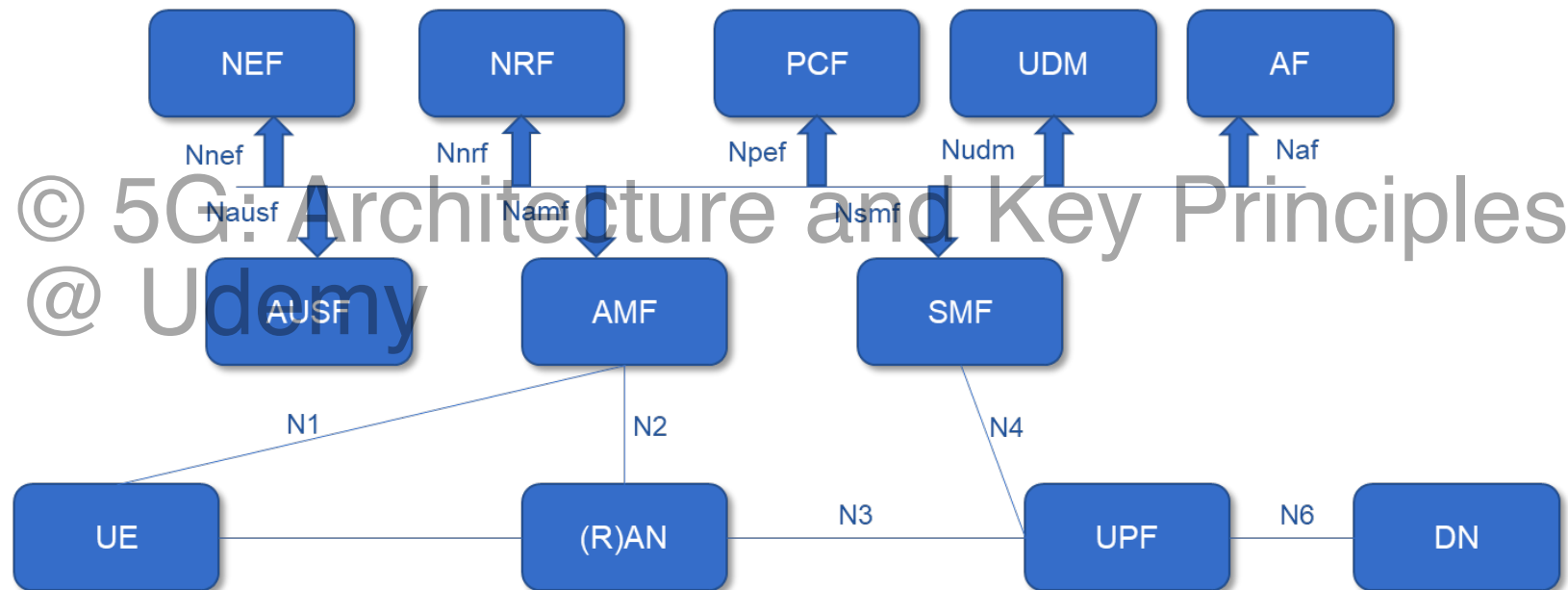


Reference Point Representation



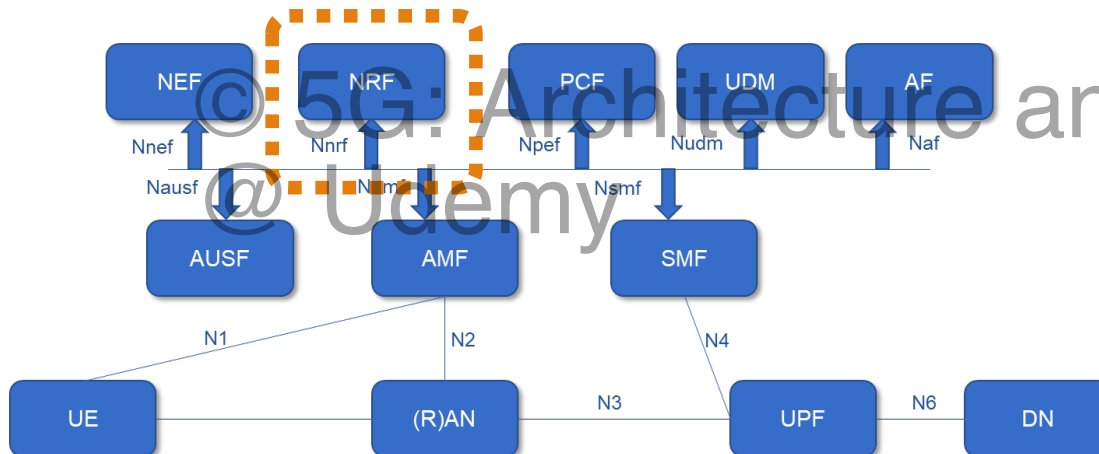
Interfaces are shown as point to point connections.

Service-Based Representation



Network function shown as services. Core network Interfaces define the service interface.

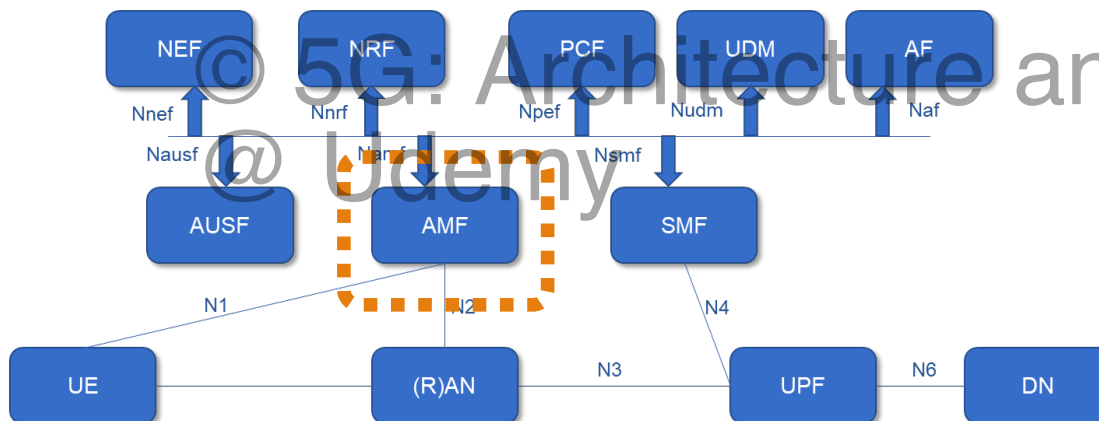
Core Network Functions – NRF



NRF (Network repository function)

- Stores the profiles of the network functions
- Service discovery

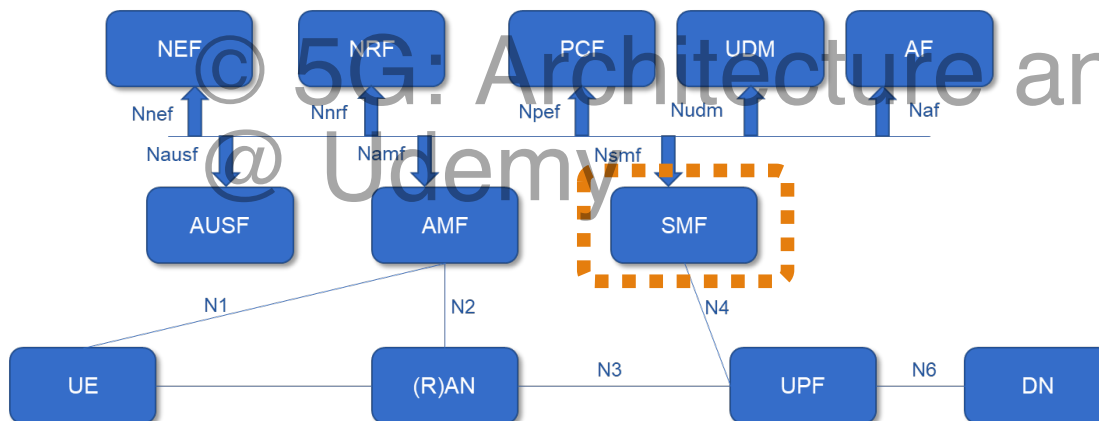
Core Network Functions – AMF



Access and Mobility Function (AMF)

- to register and authenticate UEs
- to support mobility between different radio cells

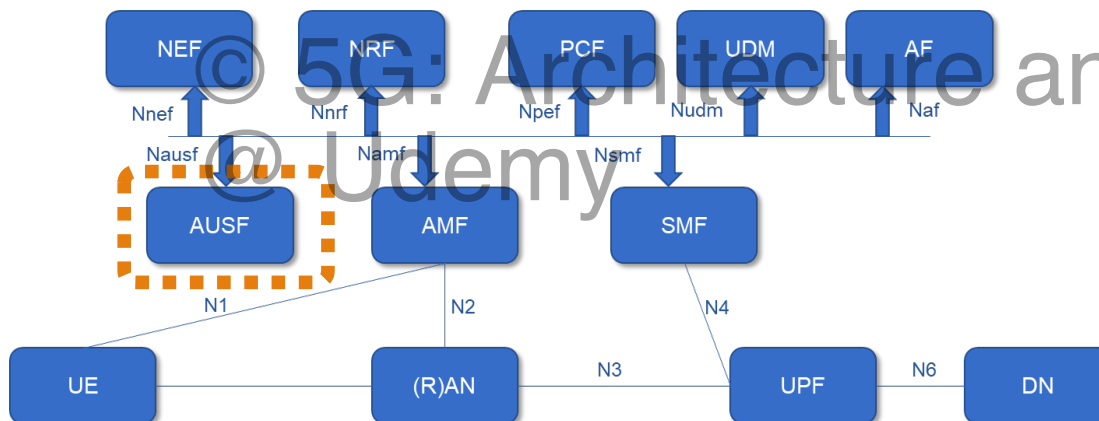
Core Network Functions – SMF



Session Management Function (SMF)

- Manages user sessions
- Selection and control of UPFs
- With PCF: UPF configuration
- With UPF: Control charging functionality

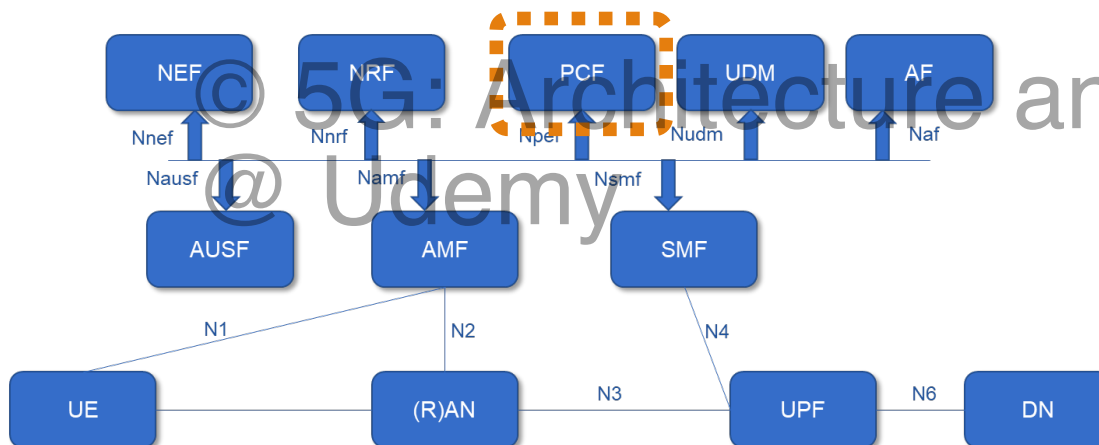
Core Network Functions – AUSF



Authentication server function (AUSF)

- Responsible for authentication, with UE and UDM

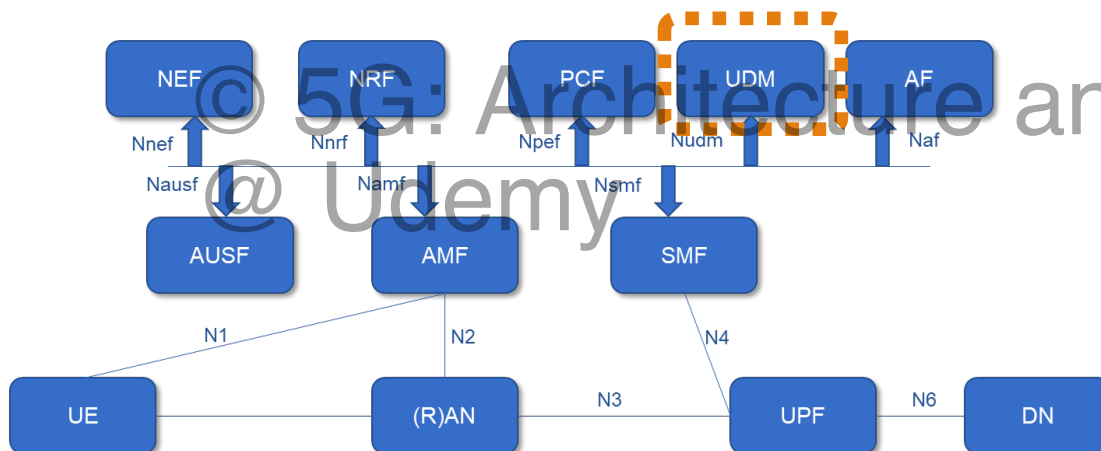
Core Network Functions – PCF



Policy Control Function (PCF)

- Session management policies
- Access and mobility policies
- UE policies

Core Network Functions – UDR / UDM



Unified Data Repository
(UDR)

- Stores subscription data

Unified Data Management
(UDM)

- Front end for subscription data

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